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Strain modulated rise, reduction, and resurrection of ferromagnetic ordering and structural properties in ZnO nanostructures due to 1.2 MeV proton implantation

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ABSTRACT

We have investigated the effect of 1.2 MeV proton implantation on the structural and magnetic properties of ZnO nanostructures. The X-ray diffraction (XRD) analysis of pristine and implanted ZnO samples revealed the presence of a wurtzite hexagonal crystal structure with space group $P6_3mc$. The XRD results also showed the emergence of a secondary phase, observed by the splitting of the (201) peak, as the 1.2 MeV proton fluences were varied. Detailed XRD analysis exhibited the strain induced in ZnO samples after proton implantation, leading to an increase in crystallite size. Room-temperature magnetization studies were carried out using SQUID-VSM. A significant enhancement in ferromagnetic ordering was observed at the fluence of 50.21 pC/μm^2, as compared to pristine samples. Interestingly, a resurgence in ferromagnetic behavior was also seen at a fluence of 100.15 pC/μm^2. Temperature-dependent zero-field cooling and field cooling magnetization measurements (2–300 K) of pristine and proton implanted samples at a fluence of 50.21 pC/μm^2 indicated that no magnetic phase transition occurs in the system. Raman spectroscopy was utilized to identify defects and chemical structure changes in the ZnO samples after proton implantation. After implantation, the defect-induced peaks at 578.47 cm^{-1} became more intense, indicating the evolution of strain. X-ray photoelectron spectroscopy analysis ruled out the presence of impurities and transition metal elements, indicating the formation of hydroxyl groups and Zn in the + 2-oxidation state in the ZnO matrix. Our results suggest that strain influenced the observed ferromagnetic ordering in the ZnO nanostructures after proton implantation.

1. Introduction

Nowadays, the demand for spintronic devices with improved quality and technology is increasing day by day, which has been fulfilled by Dilute Magnetic Semiconductor (DMS) materials to a larger extent [1–3]. Many experimental and theoretical studies have been performed to realize Room Temperature Ferromagnetism (RTFM) induced by the incorporation of dilute magnetic ions in mostly II-VI and III-V semiconductors [4–10]. Among II-VI and III-V semiconductors, ZnO is the most promising material for room temperature DMS applications, due to its theoretical prediction about the higher Curie temperature (T_c > 300 K). Additionally, ZnO is a multifunctional material having a large excitation binding energy of 60 meV and a wide-band gap of 3.37 eV, which also holds potential applications in photovoltaic devices [11], optoelectronic devices etc. [12]. Transition metals like Fe [13], Ni [14], Co [15], Mn [16] and Cu [17] are just a few examples of the magnetic elements that have been doped into ZnO semiconductors to realize DMS materials. The addition of these magnetic impurities changes the electronic, optical and magnetic properties of the ZnO semiconductors [18–20]. This produces a material that can control both charge and spin degree of freedom of electrons, making it appropriate for spintronic applications [21–23]. It has been reported that by the doping of transition metals elements in ZnO, various kinds of magnetic ordering have been established, such as ferromagnetism [16,24,25],

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anti-ferromagnetism [24,26], paramagnetism [25,27], super-paramagnetic [27,28] and spin-glass behavior [24,26], depending on the type of ions, sample shape, size and synthesis processes. Furthermore, there are varying reports from experimental studies suggesting that similar-grown samples, could exhibit paramagnetic or ferromagnetic ordering at room temperature [16,29]. It is believed that the doping of transition metals in ZnO produces the secondary phases and metallic clusters, which have deleterious effect on the applications of DMS. Above all, the major challenge has been the reproducibility and repeatability of the results from the chemical doping of transition metal elements. In addition to this, room-temperature ferromagnetic ordering has also been reported in pristine ZnO systems [30–34]. It has been reported that various mechanisms, such as the intrinsic/inherent ferromagnetism in the host material, defects, and substrate proximity effects etc. are responsible for the origin of RTFM in a pure ZnO system [30–34].

Apart from chemical doping of transition metal elements in ZnO, ion beam implantation is an alternate route to incorporate these ions into the ZnO system at specific positions of the sample in a controlled manner. It presents a variety of avenues to create desired and tailor-made controlled defect densities in the sample. The ion beam implantation of ZnO system can create a wide range of defects to a precise depth and concentration, as compared to direct doping of ions, and these defects modify the structural, optical, and magnetic properties of the material. Ion beam implantation has a great advantage over the chemical doping of ions in the host material in terms of reproducibility and repeatability. The origin of RTFM in ZnO semiconductors implanted by various transition metal ions has been probed [35–42]. It has been observed that several mechanisms, such as defects (vacancy, substitutional, interstitial), free charge carrier-mediated exchange interaction, bound magnetic polaron model, etc., have been reported for ferromagnetic ordering in ZnO host matrix implanted by a variety of transition metal ions. Room temperature ferromagnetism in ZnO has also been induced through heavy ion implantation using species such as Ni, Ar, and Xe [43–45]. These ions can create not only high defect densities and lattice strain due to their greater mass and nuclear stopping power, but may also cause amorphization or lead to the formation of secondary phases and bound magnetic polarons. These inconsistencies in the magnetic ordering as well as their mechanisms, have drawn the attention of the research community to explore the role of non-magnetic ions and elements in mediating magnetic behavior of non-magnetic ZnO semiconductors. Thereafter, RTFM has also been explored with the irradiation/doping of non-magnetic ions in ZnO by various groups. Experimental and computational studies by Pan et al. [46] have claimed, that the origin of RTFM in carbon-doped ZnO films, is attributed to the formation of the Zn-C bonds in the host material. Yi et al. [47] have shown, both experimentally and theoretically, that the Li-doped ZnO produces stable cation vacancies in the form of Li substituted at the zinc site (Li_{Zn}) and zinc vacancies (V_{Zn}), which can lead to ferromagnetism in the system. Ali et al. [48] have suggested the onset of RTFM in Ag-doped ZnO, in the framework of the bound magnetic polaron model, which is closely related to defects such as V_{Zn} responsible for the enhanced ferromagnetic ordering, while oxygen vacancies (V_o) suppress the ferromagnetism. Recently, Singh et al. [49] reported that defects such as V_o and Zn-C complex structures, along with strain, play a crucial role in the origin of RTFM as a consequence of 600 keV carbon ion irradiation into ZnO thin film systems.

Hydrogen is a versatile element present everywhere on earth and has a strong reactivity with ZnO semiconductors [50]. It has been predicted that hydrogen in ZnO possesses two types of defects: (i) interstitial hydrogen (H_i) [51] (ii) substitutional hydrogen at the O site (H_o) [52] and, (iii) formation of OH dangling bond due to hydrogen chemisorption [51]. It has also been suggested that annealing can erase the interstitial hydrogen (H_i) defects due to its high mobility [53], which indicates that the charge carrier concentration governed in the ZnO system is mostly controlled by the substitutional hydrogen H_o. The incorporation of hydrogen in the ZnO matrix can enhance the carrier-mediated exchange interaction among the localized magnetic moments due to the presence of intrinsic defects in ZnO. This has led research communities to explore the defect-induced RTFM in hydrogenated ZnO systems. The theoretical prediction by Sanchez et al. [54] states that the adsorption of hydrogen atoms on the polar ZnO (0001) surface forms H-Zn bonds, which create a metallic surface with a net magnetic moment. The origin of RTFM in ZnO porous microspheres has also been probed, and it has been suggested that defects such as zinc vacancies and shallow donors, trigger the observed ferromagnetism induced by hydrogen [55]. Li et al. [56] have reported RTFM on the surface of hydrogenated ZnO film due to the presence of unpaired electron spin in the O 2*p* orbital. Khalid et al. [57] have established RTFM at the surface of the hydrogen-plasma-treated ZnO single crystal, and it has been suggested that the observed ferromagnetic ordering is closely related to the influence of hydrogen. In addition to this, Khalid and Esquinazi [58] reported ferromagnetism in low-energy H^+ (proton) implanted ZnO crystals near the surface region (20 nm), where a significant enhancement in the saturation magnetic moment corresponding to increasing hydrogen concentration was observed. It has been suggested that the enhanced saturation magnetic moment is associated with hydrogen concentrations. The origin of RTFM in hydrogenated ZnO nanoparticles has been ascribed to the V_{Zn} and OH bonding complex (V_{Zn}+OH) [59]. Ahmed and Senthilkumar [60] observed the formation of complex clusters such as $V_{Zn}-H$, which induced the observed ferromagnetic behavior in unintentionally hydrogen-doped ZnO nanoparticles. Recently, Singh et al. [61] reported that the enhanced ferromagnetic order in the ZnO film annealed in hydrogen atmosphere. It has been suggested that the observed ferromagnetic nature is associated with the increased carrier-mediated exchange interaction among the localized magnetic moment induced by zinc vacancies along with O-H concentrations. However, the reported value of the saturated magnetic moment is quite low for practical applications in the field of spintronic devices. The origin of ferromagnetism in the ZnO system has been explained based on zinc and oxygen vacancies and some other complex defects. The previous studies have mainly focused on the observation of magnetic behavior on the surface of ZnO samples contaminated by hydrogen, where the studies inside the target depth (bulk) are broadly missing. Additionally, it has also been noticed that there is a lack of experimental and theoretical studies on the possibility of strain-induced magnetic ordering as a result of proton implantation. Proton implantation primarily generates point defects in a controlled manner through electronic energy loss, causing minimal structural disruption. This enables a more precise investigation of intrinsic defect-induced magnetism. Our study thus complements previous heavy ion work by using protons as a cleaner, less invasive tool for defect engineering in ZnO, with relevance to device applications requiring minimal lattice damage.

In this work, we have investigated the room-temperature ferromagnetic behavior inside the ZnO samples due to 1.2 MeV proton ($^1H^+$) implantation. We have attempted to correlate the observed RTFM with strain, induced by proton implantation and the subsequent formation of complex structures in ZnO samples. The detailed experimental procedure and the results will be discussed in the subsequent sections.

2. Experimental details

Commercially available high-purity ZnO polycrystalline powder (99.999 % pure, Sigma-Aldrich, USA) was utilized to prepare the samples in the form of pellets. The ZnO powder was first mixed mechanically in a mortar-pestle in wet medium (acetone) for 2 h. It was then calcined at an optimized temperature of 500 °C for 5 h in a high-purity alumina crucible (99.8 % pure, Lark, UK) in a high-temperature furnace for achieving homogeneity and to remove any residual or adsorbed impurities, if any. The calcined powder was grinded in a mortar-pestle and mixed with a small amount of binder PVA (polyvinyl alcohol) for pelletization, in the form of a 10 mm circular disc and 1000-micron thickness, utilizing the hydraulic machine under the pressure of 5.88 MPa. The prepared pellets were sintered at a temperature of 500 °C for 5 h. It has been reported that Mn-doped ZnO samples sintered at 500 °C, exhibit RTFM [16,62,63] and the observed RTFM decreases with sintering above 500 °C [16,63]. This was the basis for choosing calcination and sintering temperatures at 500 °C in the present experiment. For proton implantation and subsequent measurements, the pellets of a circular disc of diameter 10 mm were diced into a cuboid structure of 6 mm × 4 mm × 0.5 mm. The samples were pasted onto the SiO_2/Si substrate with the help of GE varnish. The purpose of pasting the samples to the SiO_2/Si substrate was to impart a minimum stress while loading and removing the samples from the sample ladder and during proton implantation, so that they do not get damaged or cracked. The ZnO samples were implanted with 1.2 MeV proton ($^1H^+$) at room temperature to four different fluences: 10.43 pC/μm^2 or ~6.52 × 10^{15} ions/cm^2, 50.21 pC/μm^2 or ~3.14 × 10^{16} ions/cm^2, 76.55 pC/μm^2 or ~4.78 × 10^{16} ions/cm^2 and 100.15 pC/μm^2 or ~6.26 × 10^{16} ions/cm^2. The beam spot size was adjusted to 5 mm × 4 mm. Based on this, a 5 mm × 4 mm area of the ZnO samples was exposed to proton beam, and the beam spot size was calibrated prior to implantation. The ZnO samples were mounted on the sample ladder using a double-sided vacuum compatible copper tape for proton implantation, maintaining chamber pressure of about 2 × 10^{-6} mbar.

The proton implantation experiments were carried out at 1.7 MV Tandetron particle accelerator (High Voltage Engineering, Europa, BV) laboratory at IIT Kanpur. The 1.2 MeV protons were extracted from a duo plasmatron source filled with hydrogen gas, with the terminal voltage of Tandetron accelerator fixed at 600 kV. The beam current of 1.2 MeV proton ions exposing the sample was kept in the range of 50–100 nA to avoid any possible heating effect on the sample during implantation. The beam current was measured using current integrator (ORTEC Model 439). In order to accurately estimate the beam current on the sample, a strong outward positive electric field was applied to the sample ladder in order to suppress the secondary electrons being emitted from the sample.

Powder X-ray diffraction (PXRD) studies of pristine and proton implanted ZnO samples were recorded using an Empyrean, Panalytical X-ray diffractometer with Cu K_α (1.5406 Å) radiation operating at 45 kV and 40 mA. The X-ray diffraction (XRD) data of pristine and implanted samples were acquired over Bragg's angle in the range of $30^\circ \leq 2\theta \leq 70^\circ$. The XRD measurements were carried out with a step size of 0.08° and a scan rate of 4° per minute. This study was essential in determining the crystal structure and phase purity of the samples before and after proton implantation. The penetration depth of X-rays in ZnO, calculated for Cu K_α radiation (λ = 1.5406 Å), is approximately 12.85 μm. The 1.2 MeV proton-induced damaged layer extends up to 11.89 μm in ZnO, as per SRIM simulations. Given the close match between the X-ray penetration depth and the predicted damage range, it can be concluded that most of the diffracted signal originates from within the implanted (damaged) region of the ZnO sample. Therefore, the contribution from the undamaged part of the bulk ZnO sample, is negligible in our measurements. To correlate the structural and chemical changes along with defects induced by proton implantation, room-temperature Raman measurements were performed using a micro-Raman spectrometer (STR-750) with an Ar-ion laser as excitation source, having excitation wavelength of 514.5 nm. The estimated penetration depth (δ) of the 514.5 nm Ar-ion laser is between 0.975 and 2.05 μm, indicating that the Raman laser probes approximately the top 1–2 μm of the ZnO sample. In order to check the room-temperature magnetization behavior of pristine and proton implanted samples, the magnetization measurements were performed using the 7 T SQUID-VSM (Quantum Design MPMS System). The ± 3 Tesla of magnetic field was applied throughout the magnetization experiments. A non-magnetic quartz sample holder was utilized for the SQUID measurement. The samples were carefully handled to avoid any foreign contamination during multiple implantation and analytical measurements. X-ray photoelectron spectroscopy (XPS) measurements were performed on the samples, using the PHI 5000 Versa Probe II, for a detailed examination of the elemental composition, impurities, and oxidation states of atoms after proton implantation. To avoid photopolymerization and moisture-related damage, the samples were moved from the accelerator laboratory to all measurements laboratories using covered vacuum desiccators.

3. Results

3.1. X-ray diffraction (XRD) analysis

The room-temperature XRD data of pristine and 1.2 MeV proton implanted ZnO samples at the fluence range of 10.43–100.15 pC/μm^2 are shown in Fig. 1. All the peaks of the XRD patterns of the pristine sample were indexed using JCPDS card no. 36–1451, and indicated the presence of a wurtzite hexagonal phase having space group $P6_3mc$ [49, 59]. It is evident from Fig. 1 that the samples implanted to fluences of 10.43 pC/μm^2, 50.21 pC/μm^2, and 100.15 pC/μm^2 have strong directional growth along the (201) plane, while pristine and the sample implanted with the fluence of 76.55 pC/μm^2, are highly oriented along the (101) plane. The intensity of (201) peak successively decreases for proton fluences up to 76.55 pC/μm^2, though the intensity of these defects induced peaks are still significant than that for the pristine sample of the (201) plane, which exhibits the incorporation of proton into the ZnO lattice.

Fig. 2(a)-(d) represents the zoomed version of XRD patterns of pristine and 1.2 MeV proton implanted ZnO samples to the fluence range of 10.43–100.15 pC/μm^2. It is observed from Fig. 2(a)-(d) that the after proton implantation peaks for planes (100), (002), (101), (102), (110), (103), (200), and (112) shift towards a higher angle, while the peaks for the plane (201) shift towards a lower angle and further splits into two peaks at the fluences of 10.43 pC/μm^2, 50.21 pC/μm^2, and 100.15 pC/μm^2. Interestingly, it is also observed from Fig. 2(a)-(d) that at the fluence of 76.55 pC/μm^2, all the peaks related to all the planes, shift towards the lower angle, in addition to this the peak for (201) plane splits into three peaks. These shifts in peak positions towards higher and lower angles indicate compressive and tensile strain, respectively, induced in the samples as a result of 1.2 MeV proton implantation [49,64]. The splitting of the (201) peak could be attributed to the formation of a functional group (OH) due to hydrogen chemisorption.

To probe the detailed structural analysis and its correlation with the magnetism, we have estimated crystallite size and lattice strain using the

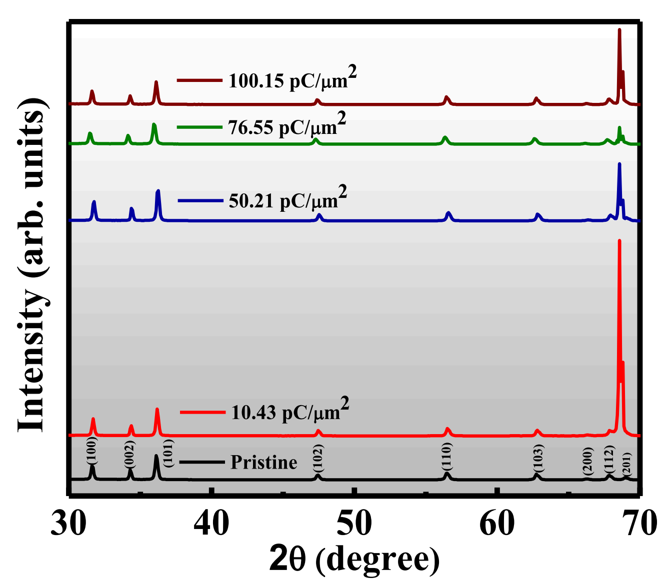

Fig. 1. The XRD patterns for pristine and 1.2 MeV proton implanted ZnO pellets to the fluence range of 10.43–100.15 pC/μm^2.

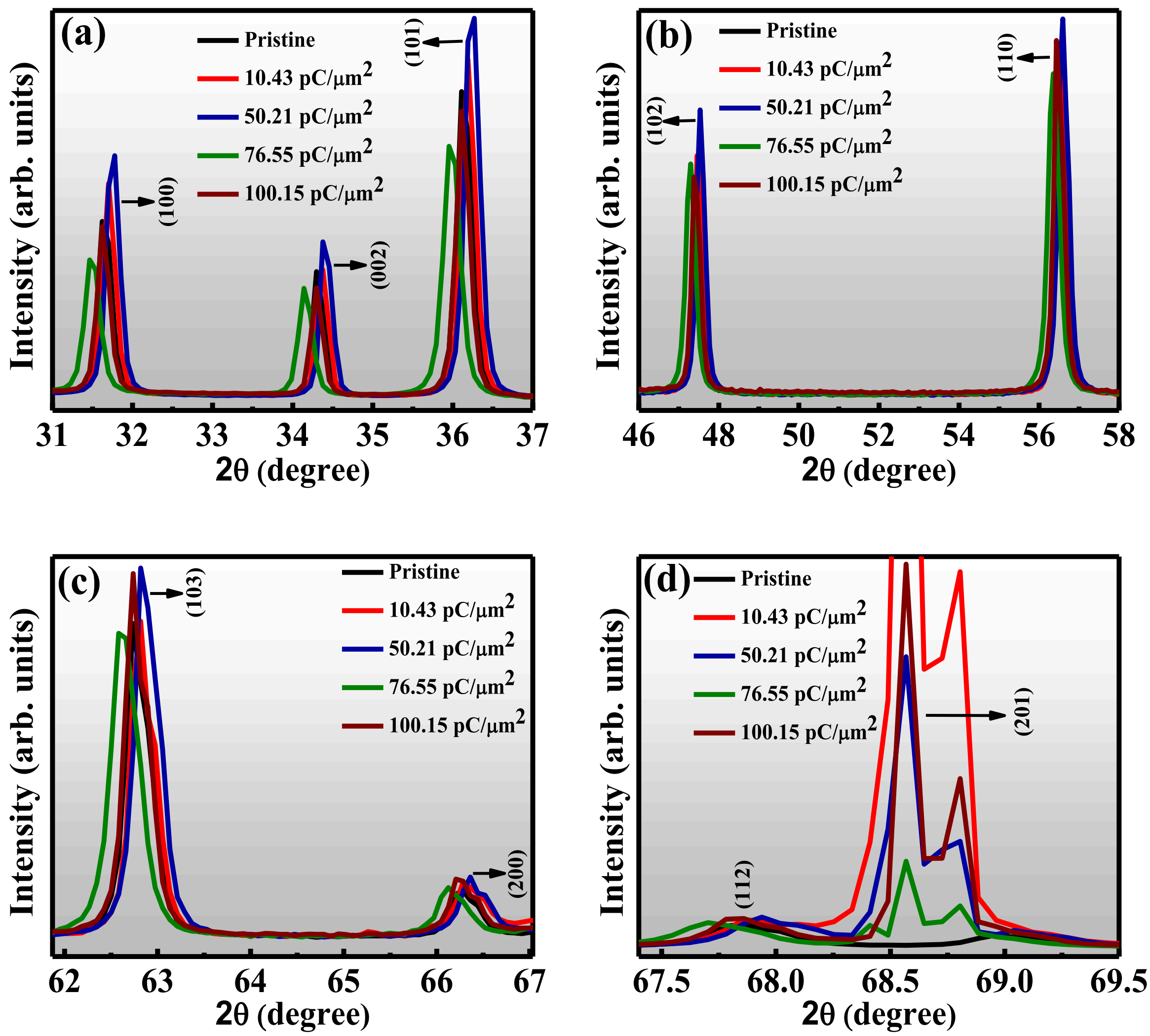

Fig. 2. Zoomed version of the XRD patterns for pristine and 1.2 MeV proton implanted ZnO pellets to the fluence range of 10.43–100.15 pC/μm^2 (a) for (100), (002), and (101) planes; (b) for (102) and (100) planes; (c) for (103) and (200) planes; and (d) for (112) and (201) planes.

Williamson-Hall (W-H) plot. In XRD patterns, the total broadening (β_T) of the peaks is due to the combined effect of crystallite size (β_D) and lattice strain (β_ε) developed inside the crystallites. The total broadening β_T can be expressed as a sum of broadening due to crystallite size and broadening due to strain in the following equations:

$$\beta_T = \beta_D + \beta_\varepsilon \tag{1}$$

$$\beta_T = \frac{k\lambda}{D\cos\theta} + \varepsilon(4\tan\theta) \tag{2}$$

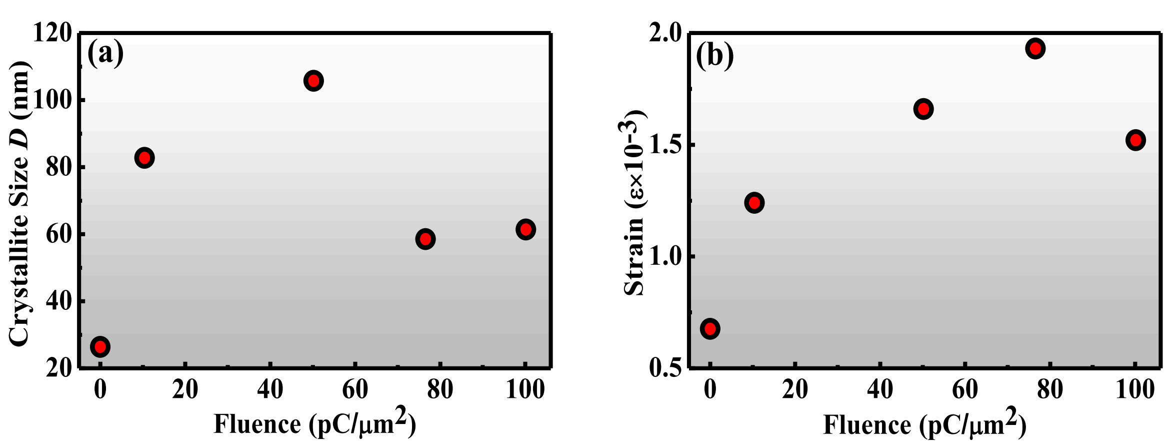

Fig. 3. (a) Variation of crystallite size in ZnO nanostructures & (b) Variation of lattice strain in ZnO nanostructures, as a function of pristine and 1.2 MeV proton fluences range of 10.43–100.15 pC/μm^2.

where, k is the Scherrer's constant equal to 0.94, λ is the wavelength of the incident X-ray ($\lambda = 1.5406$ Å), β_T is the total broadening of the peak, i.e., the Full Width at Half Maximum (FWHM), D represents the average crystallite size, θ is the Bragg's angle in radian, and ε denotes the lattice strain. The above Eq. (2) can be rearranged in the following manner:

$$\beta_T \cos\theta = \varepsilon(4\sin\theta) + \frac{k\lambda}{D} \tag{3}$$

Eq. (3) is known as the Williamson-Hall equation. The intercept and slope of the linear fitting of $\beta_T\cos\theta$ *vs* $4\sin\theta$ ($4\sin\theta$ and $\beta_T\cos\theta$ on x- and y-axes respectively) reveal the crystallite size and lattice strain respectively. As the 1.2 MeV proton fluences were varied, the corresponding variation in the crystallite size and lattice strain were assessed by the W-H plot, as shown in Fig. 3(a) & (b) respectively. It is observed from Fig. 3 (a) that the crystallite size first increases up to the fluence of 50.21 pC/μm^2, and then it decreases for the fluence of 76.55 pC/μm^2. When the fluence is further increased, the crystallite size again slightly increases. Fig. 3(b) displays an increase in the value of lattice strain, up to the fluence of 76.55 pC/μm^2, and then a decrease in strain as the fluence was further increased, indicating that the tensile strain was induced at the fluence of 76.55 pC/μm^2, and that, for all other fluences, compressive strain developed in the sample post 1.2 MeV proton implantation. This non-monotonic variation of lattice strain as a function of proton fluence, could be attributed to the trapped atoms shifting from non-equilibrium to equilibrium positions at higher fluences [65]. At the fluence of 76.55 pC/μm^2, the maximum tensile strain is induced, which signifies that at this fluence the maximum distortion occurs in the lattice which leads to reduction in the crystallite size as well, and this is the reason that why at this fluence all the peaks shifted to a lower angle as compared to that of the other fluences. The reduction in strain at the highest fluence may be attributed to the dissipation of electronic energy loss (S_e), which leads to localized thermalization and enhanced atomic mobility. These conditions can facilitate relaxation mechanisms such as defect recombination and localized atomic rearrangement, thereby reducing lattice strain despite the continued exposure to the proton beam.

Furthermore, we calculated the lattice parameters a and c by considering (100) and (002) planes, respectively. For a wurtzite hexagonal crystal structure, the relation between interplanar spacing d, lattice parameters a, c and the Miller indices (hkl) is given by:

$$\frac{1}{d^2} = \frac{4}{3}\left(\frac{h^2 + hk + k^2}{a^2}\right) + \frac{l^2}{c^2} \tag{4}$$

The following equations were utilized to assess the lattice parameters a and c, corresponding to (100) and (002) planes respectively:

$$a = \frac{\lambda}{\sqrt{3}\sin\theta} \tag{5}$$

where, λ is the wavelength of the incident X-ray ($\lambda = 1.5406$ Å) and θ is the Bragg diffraction angle corresponds to (100) plane and:

$$c = \frac{\lambda}{\sin\theta} \tag{6}$$

where, θ is the Bragg diffraction angle corresponding to (002) plane and λ is the X-ray wavelength ($\lambda = 1.5406$ Å). As observed from Fig. 4(a) the estimated values of lattice parameters a and c of pristine, as well as samples implanted with different fluences are slightly higher as compared to the standard values of JCPDS card no. 36–1451, which are listed in Table 1. It is also observed that the after implantation, lattice parameters a and c first decrease for the fluences 10.43 pC/μm^2, 50.21 pC/μm^2, and then there is a sharp increase for the fluence of 76.55 pC/μm^2, and further decrease for the fluence of 100.15 pC/μm^2. The decrease in the lattice parameters a and c after proton implantation, indicates that the compressive strain has been induced in the samples [64] and an abrupt increase in lattice parameters a and c, which takes place at the fluence of 76.55 pC/μm^2, reveals that at this fluence maximum distortion occurs in the sample, which can be observed from Fig. 3(b).

The biaxial strain model was utilized to deduce the in-plane stress (σ), given by:

$$\sigma = 4.5 \times 10^{11}\ [(c_0 - c)/\ c_0]\ N/m^2 \tag{7}$$

where, c is the calculated lattice parameter corresponding to the (002) plane in the XRD patterns and c_0 is the corresponding bulk value (5.20661 Å). It can be observed from Fig. 4(b) and Table 1, that all the samples exhibit compressive stress across the implanted fluence range [66]. Although the stress value at a fluence of 50.21 pC/μm^2 appears very close to zero in the figure, it is in fact slightly negative (–0.00090 GPa), indicating a relatively small but a measurable compressive stress. In addition, it has also been observed that the sample implanted with the fluence of 76.55 pC/μm^2, has maximum stress (-3.34738 GPa), which results in significant deformation as depicted in Fig. 3(b). The negative sign in the stress value denotes only that the stress is compressive in nature.

As different fluences of 1.2 MeV proton implantation induce a change in the lattice parameter, a change in bond length as well as unit cell volume can also be expected. Therefore, we have estimated unit cell volumes (V) for a hexagonal crystal structure and bond lengths (L) of Zn-O utilizing the following equations:

$$V = 0.866 \times a^2 \times c \tag{8}$$

$$L = \sqrt{\frac{a^2}{3} + (0.5 - u_p)^2 c^2} \tag{9}$$

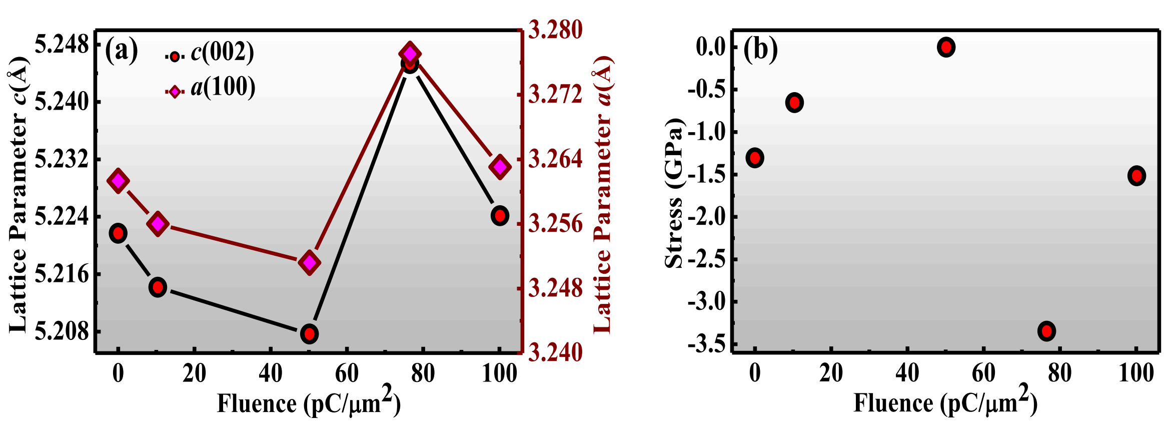

Fig. 4. (a) the variation of lattice parameters "a" and "c" & (b) the change in the residual stress, in ZnO nanostructures as a function of the pristine and 1.2 MeV proton fluence range of 10.43–100.15 pC/μm^2.

Table 1.
The table includes the standard values along with the estimated values of crystallite size, lattice strain, lattice parameter, stress, bond length, unit cell volume, degree of distortion, interplanar spacing, and dislocation density, as obtained from XRD data for the pristine sample and those implanted with different fluences.

Samples	D (nm)	ε	a(100) (Å)	c(002) (Å)	σ (002) (GPa)	L (Å)	V (Å^3)	R	d(100) (Å)	d(002) (Å)	δ lines/m^2
Standard values	-	-	3.24982	5.20661	-	-	-	-	2.81430	2.60330	-
Pristine	26.38	0.676×10^{-3}	3.26134	5.22170	−1.30421	1.984	48.13	1.01993	2.82448	2.61085	1.437×10^{15}
Implanted with 10.43 pC/μm^2	82.75	1.24×10^{-3}	3.25600	5.21418	−0.65426	1.981	47.90	1.01972	2.81986	2.60709	1.460×10^{14}
Implanted with 50.21 pC/μm^2	105.71	1.66×10^{-3}	3.25120	5.20765	−0.00090	1.979	47.70	1.01950	2.81570	2.60383	8.942×10^{13}
Implanted with 76.55 pC/μm^2	58.52	1.93×10^{-3}	3.27706	5.24534	−3.34738	1.994	48.81	1.02022	2.83810	2.62267	2.920×10^{14}
Implanted with 100.15 pC/μm^2	61.36	1.52×10^{-3}	3.26305	5.22415	−1.51596	1.985	48.20	1.01998	2.82596	2.61207	2.656×10^{14}

where a and c are the lattice parameters and u_p is the positional parameter, which can be computed by:

$$u_p = \frac{a^2}{3c^2} + 0.25 \tag{10}$$

It can be observed from Fig. 5(a) that the values of Zn-O bond length decrease for the fluences 10.43 pC/μm^2 and 50.21 pC/μm^2, indicating the shortening of the distance between Zn and O and therefore unit cell volume has also decreased. The slight decrease in the bond length of Zn-O is due to a slight decrease in the lattice parameters, and this change in the bond length is induced by the lattice strain in the ZnO systems. In case of unit cell volume assessment, a very similar trend was observed, i.e., after implantation, the unit cell volume decreases for the fluences 10.43 pC/μm^2 and 50.21 pC/μm^2. The decrease in unit cell volume reveals that the lattice shrinkage results in the formation of lattice defects in the ZnO system. An abrupt increase in the unit cell volume and bond length at the fluence of 76.55 pC/μm^2 indicates the expansion of lattice due to the induction of maximum tensile lattice strain at this fluence. The shrinkage and expansion of the lattice exhibit the consequences of incorporation of proton, which induce compressive and tensile lattice strain in the ZnO system, respectively, along with the possibility of formation of some complex structures such as Zn-OH/Zn-H etc.

The degree of distortion (R) was estimated using the following relation, to determine how much distortion occurs as a function of 1.2 MeV proton fluence in the ZnO system:

$$R = \frac{2a(2/3)^{1/2}}{c} \tag{11}$$

where a and c are the estimated lattice constants. The value of R for an ideal wurtzite structure, having no distortion in the system, is 1 [64]. In the present case, as prepared pristine and proton-implanted samples have R values between 1.01993 and 1.02022, as depicted in Fig. 5(b) as well as in Table 1. This change in R value is due to induced strain, which represents the distortion in the cell, with an increase in the proton fluence.

Interplanar spacing was estimated utilizing the Bragg's law equation,

$$n\lambda = 2d \sin\theta \tag{12}$$

where, n is the order of diffraction (usually $n = 1$), λ is the wavelength of the incident X-ray (λ = 1.5406 Å), and θ is the Bragg's angle. As observed from Fig. 5(a), the interplanar spacing for (002) and (100) planes first decrease up to the fluence of 50.21 pC/μm^2, then there is a sharp increase for the fluence of 76.55 pC/μm^2, and again it decreases for the higher fluence. It is also worth noticing that the values of interplanar spacing for (002) and (100) planes slightly increase as compared to that of the standard value of JCPDS card no. 36–1451. The slight change in interplanar spacing could be attributed to a change in lattice parameters as strain induced in the ZnO system after proton implantation.

The dislocation density (δ), which indicates the amount of defects in the sample, is described as the length of dislocation lines per unit volume of crystal and is determined utilizing the following equation:

$$Dislocation\ Density(\delta) = \frac{1}{D^2} \tag{13}$$

where, D is the crystallite size. The dislocation density decreased in all the proton implanted samples as compared to the pristine sample [49,50], as shown in Fig. 6(b). This decrease in dislocation density is a clear indication that the strain develops in the proton implanted ZnO systems.

3.2. SRIM analysis

To better understand the interaction of 1.2 MeV proton with the ZnO matrix, SRIM (Stopping and Range of Ions in Matter) calculations were

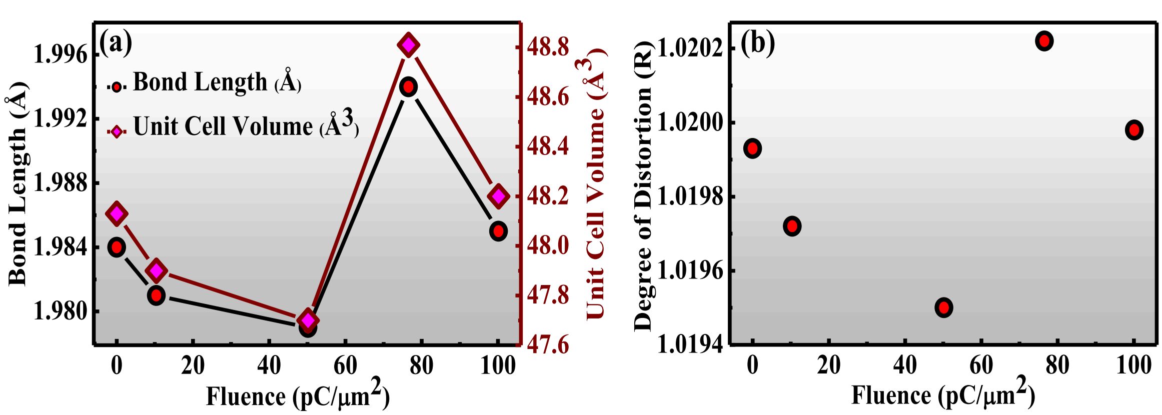

Fig. 5. Variation (a) in the bond length and unit cell volume & (b) in the degree of distortion (R), in ZnO nanostructures, corresponding to the pristine and 1.2 MeV proton fluence range of 10.43–100.15 pC/μm^2.

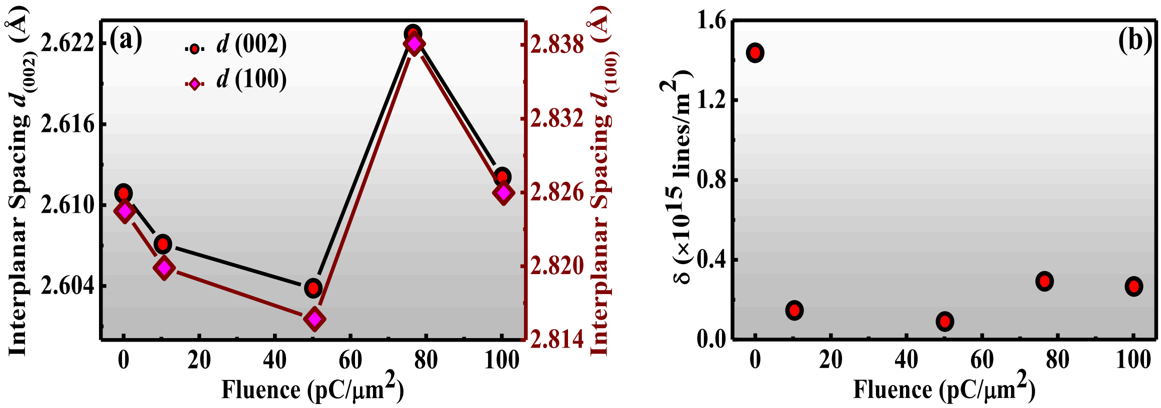

Fig. 6. Variation (a) in the interplanar spacing for the (002) and (100) planes & (b) in the dislocation density (δ), in ZnO nanostructures, corresponding to pristine and 1.2 MeV proton fluence range of 10.43–100.15 pC/μm^2.

performed [67]. The results indicate a projected range of approximately 11.89 µm, with longitudinal straggling of 6667 Å (0.6667 µm) and lateral straggling of 1.01 µm. The energy loss of protons in ZnO is dominated by electronic stopping, with an electronic energy loss of 6.813×10^{1} keV/µm, while the nuclear energy loss is significantly lower, at 5.170×10^{-2} keV/µm. This suggests that proton implantation predominantly introduces point defects via electronic interactions, without inducing significant structural damage or amorphization typically associated with higher nuclear stopping powers seen in heavy ion implantation/irradiation. As a result, proton beams offer a more controlled method for defect engineering, enabling the investigation of intrinsic defect-related phenomena in ZnO.

The depth distribution profile of 1.2 MeV proton implanted ZnO nanostructures as a function of target depth, simulated using the SRIM-TRIM code, is shown in Fig. 7. A non-uniform distribution of protons within the ZnO nanostructure is observed, with the maximum concentration occurring near the end of the projected range.

3.3. SQUID analysis

Room temperature magnetization studies of the silicon substrate, as well as pristine and implanted ZnO samples, were carried out using a SQUID-VSM magnetometer. Fig. 8 represents the measured magnetization as a function of applied magnetic field (±3 Tesla) for pristine and implanted ZnO samples without subtracting the diamagnetic contribution. The sample implanted with the fluence 50.21 pC/µm^2 responds well in terms of ion induced ferromagnetic ordering and reveals a large ferromagnetic kink as compared to that of the pristine sample. It is worth noting that the silicon substrate exhibits a strong diamagnetic nature, which eliminates any possible contribution in the magnetism from the silicon substrate.

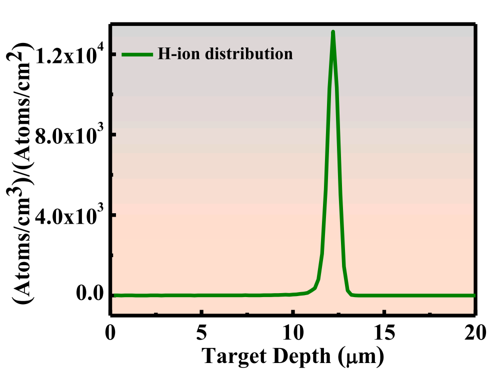

Fig. 7. Depth distribution profile of 1.2 MeV proton-implanted ZnO nanostructure as a function of target depth, simulated using the SRIM-TRIM code.

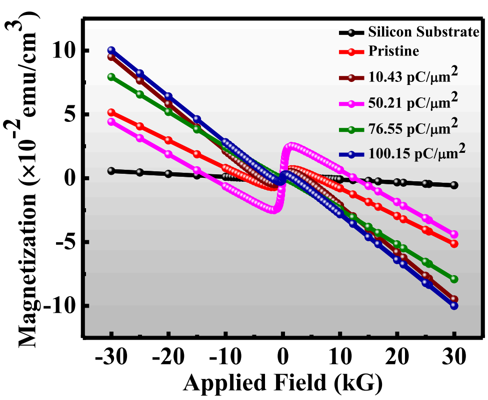

Fig. 8. Magnetization versus applied magnetic field measurements of the silicon substrate, as well as pristine and 1.2 MeV proton implanted ZnO samples at the fluence range of 10.43–100.15 pC/µm^2 without subtracting the diamagnetic contribution.

The ferromagnetic behavior of both pristine and implanted ZnO samples after subtraction of the diamagnetic contribution of the samples and bare substrate is revealed in Fig. 9. It has been ascertained that the sample implanted with the fluence of 50.21 pC/µm^2 exhibits the highest saturation magnetization (3.14×10^{-2} emu/cm^3), which is 2.31 times higher than that of the pristine sample (1.36×10^{-2} emu/cm^3). Interestingly, we also observe that the further increase in the fluence (76.55

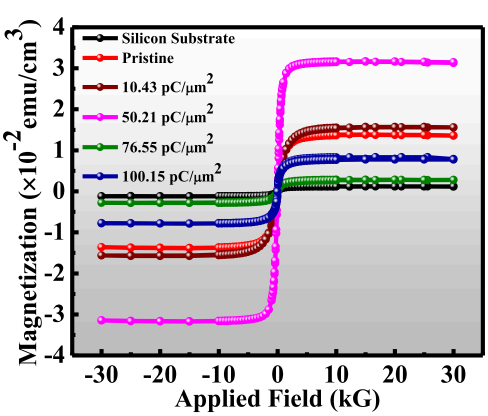

Fig. 9. Magnetization versus applied magnetic field measurements of the silicon substrate, as well as pristine and 1.2 MeV proton implanted ZnO samples at the fluence range of 10.43–100.15 pC/µm^2 after subtracting the diamagnetic contribution.

pC/μm^2) leads to a decrease in the saturation magnetization, which is less than that of the pristine sample. Again, a further increase in the fluence to 100.15 pC/μm^2 results in an increase in the magnetization value that approaches the magnetization value for the pristine sample, which indicates resurrection in the ferromagnetic behavior in the ZnO system. This result indicates that the ferromagnetic ordering increases up to a certain fluence, which defines an optimum threshold window of proton fluence. It is also noticed that the as prepared pristine sample reveals a weak ferromagnetism. It is expected that during the pelletization and sintering process some amount of strain is created, as shown in Fig. 3(b), which could be responsible for the weak ferromagnetism in the pristine ZnO sample. The significant enhancement of ferromagnetic ordering at the fluence of 50.21 pC/μm^2 may indicate the uniform distribution of protons in the ZnO system after implantation [49,68]. This uniform distribution of proton induces moderate strain as well as the possible formation of Zn-OH/Zn-H complex structures due to hydrogen chemisorption in the ZnO system, which could lead to the enhancement of saturation magnetization [54,55,69]. In hydrogen chemisorption process, hydrogen gets bonded with Zn and O molecules, which creates a dangling bond in the ZnO system that has some extra magnetic moment, and due to the presence of dangling bonds across the ZnO matrix, the system will have exchange interactions, leading to the magnetization in the material. The average magnetic moment induced by each proton ion has been estimated, and we have found that for the fluence of 50.21 pC/μm^2 proton implanted in the ZnO, each proton ion induces an average of about 5.40 μ_B of magnetic moment in the ZnO nanostructure, which is significant for practical applications in the spintronic devices. The decrease in saturation magnetization to the lowest value at the fluence of 76.55 pC/μm^2 indicates that at this fluence maximum distortion in the sample has taken place, which leads to non-uniform distribution of proton in the ZnO system, resulting in maximum tensile strain and thereby saturation magnetization decreases. The revival of the saturation magnetization, at the fluence of 100.15 pC/μm^2, could be due to moderate strain created in the ZnO matrix. Thus, our results suggest that the induction of moderate strain, along with the uniform distribution of proton ions may lead to the formation of Zn-OH/Zn-H complex structures, which plays a major role in the significant enhancement of ferromagnetic ordering in 1.2 MeV proton-implanted ZnO systems. The detailed parameters deduced are presented in Table 2.

Temperature-dependent zero field cooling (ZFC) and field cooling (FC) magnetization measurements (2–300 K) of pristine and implanted ZnO at a fluence of 50.21 pC/μm^2 are shown in Fig. 10 (a) and (b), respectively. FC–ZFC measurements were performed only for the pristine sample and the sample implanted at a fluence of 50.21 pC/μm^2, as this fluence exhibited the highest saturation magnetization. This comparison was intended to examine changes in magnetic phase between the pristine and the most magnetically enhanced sample. A continuous decrease in FC magnetization with the increase of temperature has been observed in Fig. 10 (a), except for two cusps at temperatures of ~49.3 K and ~112.3 K for the pristine ZnO sample. This monotonic decrease in FC (pristine) magnetization with temperature indicates that the frozen moments get randomized due to thermal energy. The distinct bifurcation along with a sharp increase in the magnetization at low temperatures suggests the coexistence of ferromagnetic and superparamagnetic behavior from the different regions of the pristine ZnO sample. However, our SQUID-VSM data of the pristine sample shown in Fig. 9 clearly reveals the ferromagnetic behavior at room temperature having small coercivity and saturation magnetization. The significant positive difference between the FC and ZFC curves of the sample implanted with the fluence of 50.21 pC/μm^2 shown in Fig. 10 (b), exhibits ferromagnetic ordering in the sample and eliminates any possibility of superparamagnetic and spin-glass behavior [70]. Thus, our temperature-dependent magnetization data suggests no magnetic phase transition in the ZnO system.

Table 2
The value of saturation magnetization (M_s) and the magnetic moment (μ_B) per H^+ for proton implanted ZnO nanostructure at room temperature.

Sample and nature	Conditioning	Fluence	M_s (emu/cm^3)	μ_B/H^+
ZnO pellet	1.2 MeV proton ($^1H^+$) implantation	Pristine	1.36×10^{-2}	–
		10.43 pC/μm^2	1.53×10^{-2}	1.27
		50.21 pC/μm^2	3.14×10^{-2}	5.40
		76.55 pC/μm^2	0.27×10^{-2}	0.30
		100.15 pC/μm^2	0.82×10^{-2}	0.71

3.4. Raman spectroscopy analysis

Raman spectroscopy is a highly sensitive and effective technique for analyzing changes in the local structure, defect states, and disorder within the ZnO host matrix resulting from implanted or irradiated ions. Additionally, this technique is versatile in assessing the crystalline quality of nanostructures. The hexagonal wurtzite structure of ZnO has 4 atoms in its unit cell and is associated with the symmetry space group $P6_3mc$. This space group possesses a total of 12 phonon modes, in which six are TO (transverse optical) modes, two are TA (transverse acoustic) modes, three are LO (longitudinal optical) modes, and one is LA (longitudinal acoustic) mode [64]. The optical phonons at the Γ-point of the Brillouin zone are represented by the following relation [49,64]:

$$\Gamma_{opt} = 1A_1 + 2B_1 + 1E_1 + 2E_2$$

where A_1 and E_1 modes are polar in nature, and split into TO and LO phonon modes, and all these modes are Raman and infrared active. The E_2 modes are nonpolar and divided into two modes, E_2^{high} and E_2^{low}, which are Raman active only. The B_1 modes are infrared and Raman inactive, also known as silent modes.

Fig. 11 displays the room-temperature Raman spectra of pristine and 1.2 MeV proton implanted ZnO nanostructures. It shows all the major peaks of wurtzite ZnO. All relevant Raman peaks have been identified and labeled with their corresponding vibrational modes. The peaks at ~333.2 cm^{-1} and ~381.9 cm^{-1} represent the $E_2^{(high)} - E_2^{(low)}$ and A_1 (TO) modes, respectively [59,64,71]. These peaks are attributed to the multiphonon process and indicate the single crystalline nature of ZnO. As the 1.2 MeV proton fluence is varied, the intensity of the peaks decreases, while there is no shift observed in the $E_2^{(high)} - E_2^{(low)}$ vibration mode as compared to that of pristine sample. This indicates that there is a change in the local geometry of the nanostructures, due to the incorporation of proton in the ZnO host matrix while retaining the crystal structure.

To highlight the tensile strain, a zoomed view of the Raman spectra is shown in Fig. 12. The peak centered around ~440.5 cm^{-1} corresponds to the high-frequency branch of E_2 mode (E_2^{high}) [64]. The E_2^{high} vibrational mode is due to the vibration of the oxygen sublattice, which is the characteristic and major peak of the wurtzite ZnO structure. It is observed that all the samples after implantation with different fluences of 1.2 MeV proton exhibit a fixed shift (2.0 cm^{-1}) in the nonpolar E_2^{high} mode. This fixed shift in the peak position of the nonpolar E_2^{high} mode after proton implantation, indicates the redshift, which signifies the fact that the incorporation of proton creates the atomic structural irregularities in the ZnO host matrix and disrupts its translational symmetry, resulting in an increase in local lattice alterations. This disorder and local distortion disrupts the long-range order in ZnO, which affects the mode that is related to the electric field. The pristine sample shows the most prominent peak at the frequency of ~582.44 cm^{-1}, which can be allocated to the mixture of both LO (A_1 and E_1) vibrational modes of ZnO [64]. It is observed that all the implanted samples at different fluences, show nearly a similar shift of 3.97 cm^{-1}, toward a lower wave number as

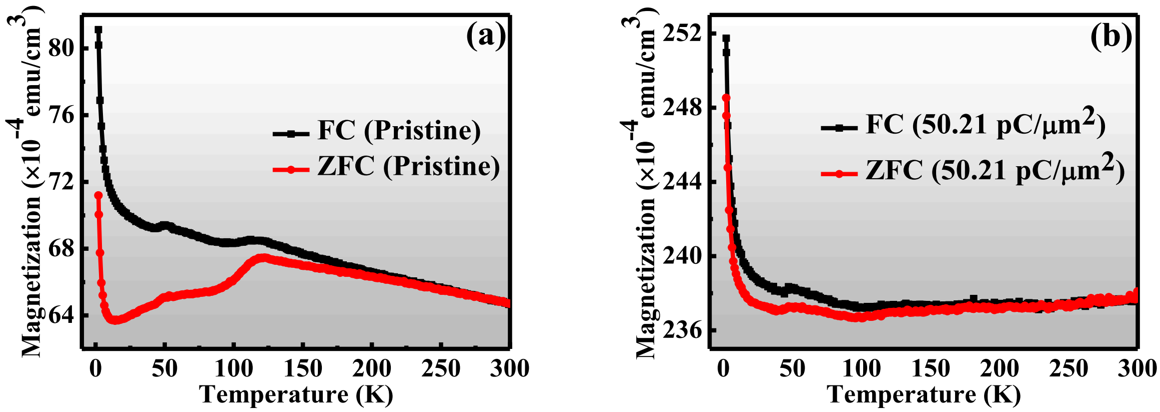

Fig. 10. Temperature-dependent zero field cooling (ZFC) and field cooling (FC) magnetization measurements in the presence of a 1000 Oe applied magnetic field for (a) pristine ZnO, and (b) 1.2 MeV proton implanted ZnO samples at a fluence of 50.21 pC/μm^2.

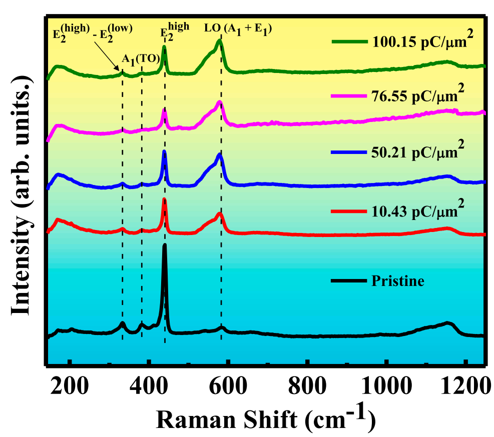

Fig. 11. Raman spectra of pristine and implanted ZnO samples at the fluence range of 10.43–100.15 pC/μm^2.

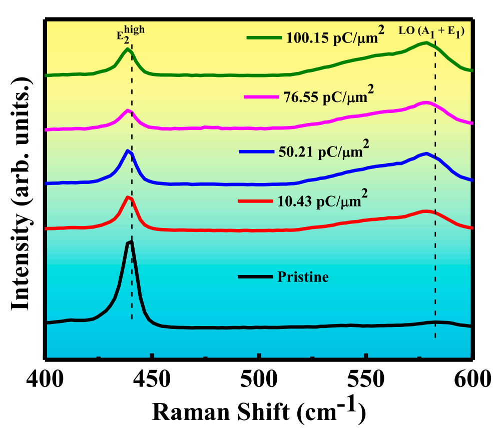

Fig. 12. Magnified Raman spectra of E_2^{high} and LO ($A_1 + E_1$) vibrational modes of pristine and 1.2 MeV proton implanted ZnO samples at the fluence range of 10.43–100.15 pC/μm^2.

compared to the pristine sample peak at ~582.44 cm^{-1}. This nearly fixed shift (3.97 cm^{-1}) in peak position toward lower wave number, is hallmark of a redshift, which indicates the presence of tensile strain [49]. As the fluence is varied, the Raman mode of the pristine sample (~582.44 cm^{-1}) not only shifts to lower wave number but also the peaks became broader and highly intense, revealing the evolution of strain in the ZnO system. Thus, our analysis of Raman spectra suggests that the defect in the form of strain is introduced, after proton implantation in the ZnO system, which results in the chemical and structural changes, and thus enhances the magnetization behavior of the ZnO nanostructures.

3.5. X-ray photoelectron spectroscopy (XPS) analysis

The X-ray photoelectron spectroscopy (XPS) study has been performed to investigate the chemical bonding states of the atoms and to identify any impurities or transition metal elements in the proton implanted ZnO nanostructures. Fig. 13 (a)-(d) exhibits the complete survey scan and binding state for the C 1s, O 1s, and Zn 2p core-level spectra respectively, for the ZnO sample implanted with the fluence of 50.21 pC/μm^2, where the observed magnetization has been maximum. The complete survey scan of ZnO sample implanted at a fluence of 50.21 pC/μm^2, indicates the presence of C, Zn, and O elements [Fig. 13 (a)]. We did not find any transition metal impurities in the implanted sample within the detection limit of XPS instrument. The detection limit of XPS instrument is typically around 0.1 at. percent, which corresponds to approximately 1000 parts per million (ppm). Thus the observed magnetization can not be pinned upon foreign impurities/contaminants.

The XPS spectra of C 1s, O 1s, and Zn 2p binding states were fitted utilizing the Voigt function. The binding scale was measured after charge correction, utilizing the C 1s peak at 284.6 eV. The XPS spectrum of carbon exhibits an asymmetric 1s peak in Fig. 13 (b), which has been deconvoluted into the three peaks. The high-intensity peak of the C 1s spectra, denoted by C_1 at a lower binding energy of ~284.6 eV, as shown in Fig. 13 (b), indicates the presence of a C=C bond. In contrast, the second peak, C_2, at ~285.8 eV, is attributed to the C-O bond, and the low-intensity peak, C_3, at the higher binding energy of ~287.5 eV, is associated with the C=O bond [72]. The core-level spectrum of O 1s shows three broad and asymmetric peaks, as illustrated in Fig. 13 (c), indicating three distinct features in the sample. The lower binding energy peak i.e. peak O_1 at ~530.3 eV is associated with the formation of Zn-O bonding state i.e. the hexagonal wurtzite structure of ZnO features O^{2-} ions which are associated with Zn^{2+} ions [49,61,72]. The second peak, referred to as O_2, appears at a binding energy of ~531.5 eV and this could indicate the presence of oxygen vacancies (V_o) [61]. In contrast, the peak labeled as O_3, at a higher binding energy of ~532.4 eV, is associated with absorbed oxygen species or the formation of hydroxyl groups (OH species) in the ZnO system, after proton implantation [49,61]. The core-level spectrum of Zn 2p states displays two distinct peaks at binding energies of ~1021.3 eV and ~1044.5 eV, corresponding to the Zn $2p_{3/2}$ and Zn $2p_{1/2}$ states, respectively, as shown in Fig. 13 (d). The difference in binding energies between the Zn $2p_{3/2}$ and Zn $2p_{1/2}$ states, i.e., the spin-orbit splitting value, is measured to be 23.2 eV, which is consistent with previously reported values [61]. This

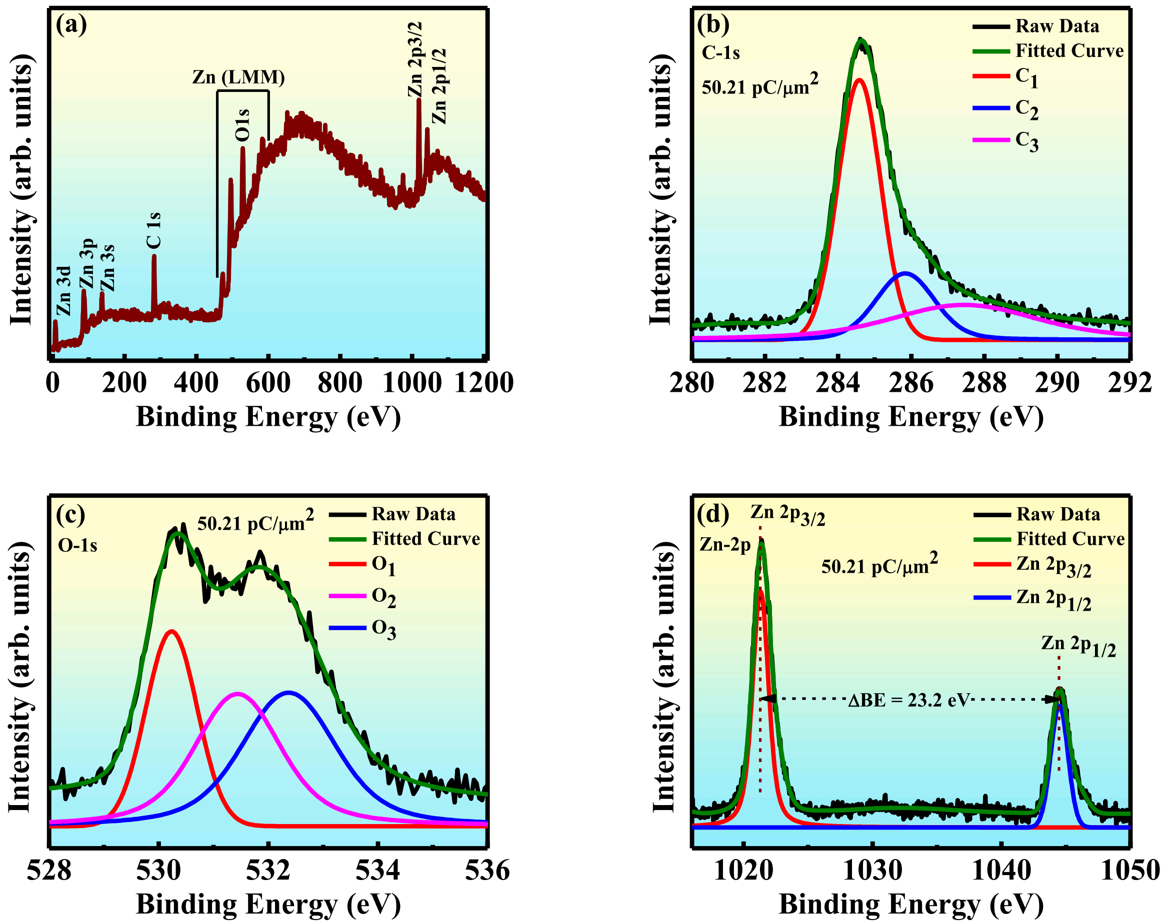

Fig. 13. (a) Wide scan XPS survey spectra, and high-resolution XPS spectra for (b) C 1s core level spectra (c) O 1s core level spectra and (d) Zn 2p core level spectra of ZnO sample implanted at the fluence of 50.21 pC/μm^2.

difference in the binding energies of Zn 2p$_{3/2}$ and Zn 2p$_{1/2}$ states indicate that all the Zn ions are in the + 2-oxidation state inside the hexagonal wurtzite structure of ZnO, which also eliminates the possibility of metallic Zn existing at the interstitial sites. Thus, our XPS results eliminate the presence of transition metal elements and any other impurities on the surface of the ZnO sample after proton implantation to the fluence of 50.21 pC/μm^2. Additionally, the presence of hydroxyl groups (OH species) leading to the formation of Zn-OH/Zn-H complex structures within the hexagonal wurtzite structure of ZnO, has been established, post 1.2 MeV proton implantation at the fluence of 50.21 pC/μm^2.

4. Discussion

The room temperature ferromagnetism in dilute magnetic semiconductors (DMS) is affected by several factors, including charge carrier concentration, types of magnetic ions, their exchange interactions, and the influence of defects and temperature, which have been previously reported. Here, our perspective was to correlate the onset of ferromagnetism with the lattice strain induced by 1.2 MeV proton implantation on ZnO nanostructures. Some previous studies have reported room-temperature ferromagnetism in hydrogenated ZnO, i.e., ferromagnetism arises only from the surface of the samples. Sanchez et al. [54] reported that H-adsorption at one monolayer leads to the formation of Zn-H bonds, which results in a hole-doped metallic surface, i.e., a fully hydrogenated surface exhibits surface magnetization with a net magnetic moment of 0.5 μ_B per hydrogen atom. In contrast, when there was only partial coverage of hydrogen, the Zn-H bonds became stronger and resulted in the diminishing of magnetism in the system. It was reported that post-hydrogenation, non-magnetic ZnO displays room-temperature surface ferromagnetism, and further annealing in a pure Ar atmosphere, leads to disappearance of the ferromagnetic ordering [56]. It was suggested that post-hydrogenation, OH-bonds are responsible for surface ferromagnetism, and subsequent annealing in Ar environment suppresses the formation of the OH-bonds, leading to the disappearance of ferromagnetic ordering [56]. This experimental result was well supported by a first-principles calculation, which indicated that the OH-terminated ZnO surface has the lowest formation energy of −2.97 eV with a net magnetic moment of 0.30 μ_B per OH-bond, due to the unpaired electron spin in the O 2p orbital [56]. Xia et al. [55] reported that the thermal decomposition of $Zn_5(OH)_8Ac_2 \cdot 2\ H_2O$ into ZnO, fabricated by annealing the precursor $Zn_5(OH)_8Ac_2 \cdot 2\ H_2O$ at 400 °C, results in the highest saturation magnetization. It was suggested that both shallow donors created by hydrogen and zinc vacancies contribute to room-temperature ferromagnetic ordering. However, further annealing of the precursor at 500 °C decreased the concentration of these defects, which subsequently reduced the ferromagnetic ordering. Khalid et al. [57] reported room-temperature ferromagnetism in hydrogen-implanted ZnO single crystals post hydrogen-plasma treatment. They suggested that only a 20 nm thick surface region contributed to the total ferromagnetic signal, with a net magnetic moment of 0.2 μ_B per ZnO unit cell, and this induced ferromagnetism in hydrogen-implanted ZnO single crystals was attributed to the influence of elevated hydrogen concentration. Khalid et al. [58] also performed a similar study, and the results suggested that the hydrogen concentration plays a major role in enhancing ferromagnetic ordering in 330 eV proton (H^+) ion-implanted ZnO single crystals. Experimental studies conducted by Xue et al. [59] suggested that the Zn vacancy and OH-bonding complex (V_{Zn}+OH) are key factors in the observed room-temperature ferromagnetism in hydrogenated ZnO nanoparticles. This finding was further supported by their computational results, which indicated that the combination of zinc vacancy and OH bond (V_{Zn}+OH) configuration has the lowest formation energy, having a magnetic moment of 0.57 μ_B.

Sub-micrometric powders of ZnO demonstrated ferromagnetic behavior, following a high-pressure hydrogen heat treatment [73]. X-ray absorption near-edge measurements indicated that hydrogen acts as a shallow donor, contributing electrons to the conduction band (Zn 4s). It was suggested that the observed magnetism was a superficial phenomenon associated with the bonding of hydrogen to zinc (Zn) or oxygen (O), on the polar surfaces of ZnO [73]. A recent ab initio study carried out by Robaina et al. [69] found, that hydrogen adsorption on the polar surface of ZnO (0001) leads to the emergence of ferromagnetism due to surface interactions. The findings indicated that with complete hydrogen coverage over the Zn atoms of the polar Zn-Zn (0001) surface, strong H-Zn bonds are formed, resulting in a net magnetic moment of 1 μ_B per unit cell. Furthermore, a fully hydrogenated ZnO monolayer with zinc vacancies (V_{Zn}) and hydrogen-passivated zinc vacancies ($V_{Zn}-H$) demonstrated half-metallic behavior, exhibiting a surface magnetic moment of 2 μ_B per unit cell [69]. E. Ahmed and K. Senthilkumar reported that formation of $V_{Zn}-H$ complex defects in unintentionally hydrogen-doped ZnO nanoparticles, were responsible for the ferromagnetic ordering [60]. However, their temperature-dependent field-cooled (FC) and zero-field-cooled (ZFC) measurements at different applied magnetic fields of 50 Oe and 100 Oe indicated the presence of various types of magnetic behavior e.g. ferromagnetic, antiferromagnetic, superparamagnetic, and spin glass characteristics. Singh et al. [61] reported that the enhancement of ferromagnetic behavior in ZnO films post-hydrogen heat treatment is associated with the $V_{Zn}-OH$ complex defects. Their experimental results were supported by theoretical finding as well, which revealed that the $V_{Zn}-OH$ configuration produces a saturation magnetic moment of 4.121 μ_B with a low formation energy.

In addition to defect complexes such as $V_{Zn}-OH$ and $V_{Zn}-H$, and the formation of Zn-H/Zn-OH bonding states and OH functional groups, which arise post hydrogen heat treatment or ion irradiation/implantation in ZnO systems, our study highlights the critical yet often overlooked role of strain in influencing magnetism. Despite its significant impact on material properties, this aspect has not been widely explored by research communities. Generally, ion irradiation/implantation is utilized to create various types of defects; however, in this case, we aim to utilize ion beam-induced defects to introduce strain into the system. To examine the effect of strain on the enhancement of ferromagnetic ordering after implantation, we performed 1.2 MeV proton implantation on the ZnO nanostructures. The detailed analysis of the XRD results indicates that the shift in peak position towards higher and lower angles corresponds to the fluences of 10.43 pC/μm^2, 50.21 pC/μm^2, 100.15 pC/μm^2, and 76.55 pC/μm^2, respectively, due to the presence of compressive and tensile strain in the system. In addition, the XRD results also elucidate the formation of a secondary phase due to the splitting of the (201) peak as the 1.2 MeV proton fluence is varied. The formation of the secondary phase may be due to the hydrogen chemisorption mechanism, as it is expected that, after proton implantation, hydrogen may bond with oxygen to form OH-functional groups, which is also well supported by our XPS results. Furthermore, the analysis of crystallite size and lattice strain reveals that the highest crystallite size and lattice strain are obtained at the fluences of 50.21 pC/μm^2 and 76.55 pC/μm^2, respectively as shown in Table 1. The highest saturation magnetic moment is achieved at a fluence of 50.21 pC/μm^2, as shown in Table 2. This indicates that a fluence of 50.21 pC/μm^2 serves as a threshold for inducing stable and controlled strain in the sample, which leads to the formation of the largest crystallite size and, consequently, enhances the saturation magnetic moment to its maximum value in ZnO nanostructures. Interestingly, the sample implanted at a fluence of 50.21 pC/μm^2 exhibits the highest magnetization, which coincides with the maximum crystallite size observed at this fluence. In contrast, the sample implanted at 76.55 pC/μm^2 shows the lowest magnetization, corresponding to the smallest crystallite size. This suggests a direct correlation between crystallite size and magnetic behavior in the ZnO system after proton implantation. A possible physical mechanism behind this trend involves the interplay between ion-induced defect formation and in-situ energy deposition during implantation. At moderate fluence, the energy transferred by the incoming protons is sufficient to activate defect migration and localized atomic rearrangements, promoting grain coarsening and the growth of larger crystallites. These larger crystallites may support enhanced ferromagnetic ordering, potentially due to more efficient alignment of defect-induced magnetic moments or reduced grain boundary scattering. In contrast, at higher fluences, either insufficient energy or excessive damage can hinder grain growth, resulting in smaller crystallites and a corresponding reduction in magnetization. This behavior emphasizes the critical role of ion fluence in tuning both the structural and magnetic properties of ZnO through implantation. Another possible explanation for the observed correlation between increased crystallite size and enhanced magnetization suggests a key role of grain boundaries in modulating the magnetic behavior of ZnO nanostructures. At moderate fluence, electronic energy loss from 1.2 MeV proton implantation can lead to localized thermal spikes, promoting grain boundary mobility and grain coalescence, thereby increasing crystallite size. This reduction in grain boundary density can suppress spin disorder and enhance ferromagnetic ordering by facilitating better exchange interactions across the grains [74]. What happens at the fluence of 76.55 pC/μm^2 that decreases the saturation magnetic moment to its lowest value, even lower than that of the pristine sample? At a fluence of 76.55 pC/μm^2, the maximum strain is induced in the sample, which deforms the ZnO lattice the most. This leads to an enhancement in the lattice parameter, while the corresponding crystallite size decreases. At this fluence, the enhanced lattice parameter results in an increase in the Zn-O bond length, as shown in Table 1, which weakens the exchange interaction between the atoms and hence causes the saturation magnetic moment to decrease to its lowest value. It is noticeable that a further increase in fluence (100.15 pC/μm^2) results in a slight decrease in strain, while the corresponding crystallite size increases. This suggests that as strain decreases, deformation in the ZnO lattice also decreases, which stabilizes the strain to some extent in the system. As a result, the saturation magnetic moment slightly increases (see Table 2), indicating the revival of ferromagnetic behavior in 1.2 MeV proton implanted ZnO nanostructures, as shown in Fig. 8. At the highest fluence, defect clustering, localized heating, and saturation effects may lead to a nonuniform distribution of protons. The observed revival of magnetization at 100.15 pC/μm^2 is possibly linked to moderate strain, as supported by XRD and Raman analysis. The dislocation density analysis is consistent with our prediction that strain is induced as the 1.2 MeV proton ion fluence varies in the ZnO system. The dislocation density reflects the number of defects present in the sample. The decrease in dislocation density as the proton fluence changes, indicates that the strain has developed in the ZnO host matrix. The shift in Raman peaks, particularly the A_1 (TO) mode, provides insight into the stress state of the ZnO nanostructures after proton implantation. With increasing proton fluence, a fixed blue shift of the A_1 (TO) phonon mode is observed, which is typically associated with the development of compressive stress within the lattice. This observation is consistent with the XRD analysis, where all the samples exhibit compressive stress, as shown in Fig. 4(b) and Table 1. The compressive nature of the stress inferred from both techniques support the conclusion that proton implantation induces lattice distortion and strain due to the accumulation of defects and interstitials. This correlation between the Raman and XRD data confirms that the strain plays a critical role in modulating the structural and magnetic properties of ZnO after proton implantation. Furthermore, the Raman spectra confirm the preservation of the wurtzite structure of ZnO, despite the lattice strain induced by implantation. The M-H curve confirms the RTFM, exhibiting coercivity and saturation magnetization. Temperature-dependent FC and ZFC measurements provide further evidence of ferromagnetic ordering, showing distinct bifurcation and irreversibility in the low-temperature region, phenomena commonly interpreted as indicative of ferromagnetism in the system. Our prediction for the induced ferromagnetism after proton implantation is as follows: It is expected that hydrogen bonds with zinc and oxygen (Zn-H-O) due to the strain generated in the system. A neutral Zn atom has filled $4s^2$ and $3d^{10}$ orbital. Since XPS studies indicate the presence of Zn in the + 2-oxidation state, this suggests that Zn loses two electrons from the 4s orbital, leaving it empty. Hydrogen, which has a half-filled 1s orbital, interacts with the Zn (4s) orbital through exchange interactions, resulting in a net magnetic moment. A similar mechanism occurs between the O 2p and H 1s orbitals, contributing to the net magnetic moment in the system. Additionally, the presence of Zn in the + 2-oxidation state eliminates any possibility of metallic Zn at interstitial sites, suggesting that hydrogen bonds with zinc in addition to oxygen.

It has been reported by Chen et al. [75] that hydrogen ions can agglomerate and form hydrogen bubbles within ZnO under certain conditions, particularly during high-dose hydrogen implantation or annealing processes. The formation of such bubbles can lead to localized lattice distortions, strain, and the creation of extended defects, which may in turn influence both the structural integrity and magnetic behavior of the material. Although no direct evidence of hydrogen bubble formation was observed in the present study due to the relatively moderate ion fluences. It is possible that nano-scale hydrogen clustering or early-stage bubble formation could contribute to the observed changes in crystallinity and magnetic response. This possibility highlights the complex role of hydrogen in ZnO and suggests that future investigations should incorporate advanced microscopic and spectroscopic methods to more clearly identify and quantify the presence of hydrogen bubbles.

5. Conclusion

The systematic room-temperature ferromagnetic ordering of pristine and ZnO implanted with protons to different fluences were studied. The XRD studies revealed a distinct phase formation in addition to the wurtzite hexagonal structure and development of strain after proton implantation. The results demonstrated that the higher crystallite size induces higher degree of magnetization, and the lower crystallite size yields lower magnetization in the ZnO system, after proton implantation. It was observed that the maximum crystallite size and the highest induced magnetization in the sample occurs at a fluence of 50.21 pC/μm^2. A significant revival of ferromagnetic behavior was also observed at a fluence of 100.15 pC/μm^2. Raman analysis also revealed that strain was induced in ZnO systems after proton implantation. XPS studies ruled out the presence of any impurities and transition metal elements in the samples. The analysis of the XPS results demonstrated the presence of hydroxyl (OH) groups, leading to the formation of Zn-OH complex structures within the ZnO host material. All the results indicated that the strain modulated the observed ferromagnetic ordering in the ZnO nanostructures after proton implantation. Our findings emphasize the need for more focused research on strain as a key contributor to magnetism in the ZnO systems. Future studies should employ advanced microscopy and spectroscopy techniques to more precisely identify and quantify defect formation. The application of proton implantation to induce room-temperature ferromagnetic ordering in ZnO nanostructures paves the way for new possibilities in spintronic device applications.

CRediT authorship contribution statement

Sheshamani Singh: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Neeraj Shukla:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation. **Paras Poswal:** Writing – review & editing, Validation, Investigation. **Rohit Tyagi:** Validation, Investigation, Data curation. **Ram Kumar:** Validation, Investigation. **Sandeep B. Bari:** Validation, Investigation. **Sujay Chakravarty:** Validation, Investigation. **Aditya H. Kelkar:** Validation, Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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